STACKS, DIFFERENTIAL GRADED SHEAVES: THEOREM-QIS-INTO-DG-INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4232-4242 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4244-4290, the complete author proof (45 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_sdga.json (pb=2255, stmt 4232-4242, proof 4244-4290).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original theorem 079F: sources/stacks-sdga/pinned-original-source.tex, source lines 4290–4290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0FSY: sources/stacks-sdga/pinned-original-source.tex, source lines 4062–4082. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0FSZ: sources/stacks-sdga/pinned-original-source.tex, source lines 4184–4206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. For every differential graded $\mathcal{A}$-module $\mathcal{M}$ there exists a quasi-isomorphism $\mathcal{M} \to \mathcal{I}$ where $\mathcal{I}$ is a graded injective and K-injective differential graded $\mathcal{A}$-module. Moreover, the construction is functorial in $\mathcal{M}$.*

Proof. Let R and $\mathcal{M}_r \to \mathcal{M}'_r$ be a set of morphisms of $Mod(\mathcal{A}, \mathrm{d})$ found in Lemma 0FSY. Let M with transformation $\mathrm{id} \to M$ be as constructed in Lemma 0FSZ using R and $\mathcal{M}_r \to \mathcal{M}'_r$. Using transfinite recursion we define a sequence of functors M_α and natural transformations $M_\beta \to M_\alpha$ for $\alpha < \beta$ by setting

(1) $M_0 = \mathrm{id}$,

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(2) $M_{\alpha+1} = M \circ M_\alpha$ with natural transformation $M_\beta \to M_{\alpha+1}$ for $\beta < \alpha + 1$ coming from the already constructed $M_\beta \to M_\alpha$ and the maps $M_\alpha \to M \circ M_\alpha$ coming from $\mathrm{id} \to M$, and
(3) $M_\alpha = \mathrm{colim}_{\beta<\alpha} M_\beta$ if α is a limit ordinal with the coprojections as transformations $M_\beta \to M_\alpha$ for $\alpha < \beta$.

Observe that for every differential graded $\mathcal{A}$-module the maps $\mathcal{M} \to M_\beta(\mathcal{M}) \to M_\alpha(\mathcal{M})$ are injective quasi-isomorphisms (as filtered colimits are exact).

Recall that $Mod(\mathcal{A}, \mathrm{d})$ is a Grothendieck abelian category. Thus by Injectives, Proposition 079F (applied to the direct sum of $\mathcal{M}_r$ for all $r \in R$) there is a limit ordinal α such that $\mathcal{M}_r$ is α-small with respect to injections for every $r \in R$. We claim that $\mathcal{M} \to M_\alpha(\mathcal{M})$ is the desired functorial embedding of $\mathcal{M}$ into a graded injective K-injective module.

Namely, any map $\mathcal{M}_r \to M_\alpha(\mathcal{M})$ factors through $M_\beta(\mathcal{M})$ for some $\beta < \alpha$. However, by the construction of M we see that this means that $\mathcal{M}_r \to M_{\beta+1}(\mathcal{M}) = M(M_\beta(\mathcal{M}))$ factors through $\mathcal{M}'_r$. Since $M_\beta(\mathcal{M}) \subset M_{\beta+1}(\mathcal{M}) \subset M_\alpha(\mathcal{M})$ we get the desired factorizaton into $M_\alpha(\mathcal{M})$. We conclude by our choice of R and $\mathcal{M}_r \to \mathcal{M}'_r$ in Lemma 0FSY. $\square$